

ZHANG Jie

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EDUCATION

North University of China

09/2015-07/2019

- **Major:** Mechanical Electronics Engineering
- **GPA:** 3.73/5.00 **Rank:** 4/146 (Top 3%)
- **Scholarship:** First-Class Scholarships (06/2019 & 12/2018) and Second-Class Scholarship (12/2017)

RESEARCH

Graduation Design: QT-based Multi-channel Temperature Acquisition and Display System

10/2018-05/2019

Author

- Designed and built a multi-channel temperature hardware system based on the MODBUS-RTU communication protocol, consisting of thermocouple temperature sensors, temperature acquisition modules, and an RS-485 bus
- Developed a QT-based upper computer application to achieve real-time temperature monitoring, historical temperature curve plotting, and data export functions
- Debugged the system, tested system performance, and deployed the QT-based application on a development board to validate system stability and operational efficiency

WORKING EXPERIENCE

Gree Electric Appliances, Inc. of Zhuhai

07/2019-Present

Team Lead of Control Technology R&D Engineer

- Designed and developed display software for a 10kg heat pump all-in-one washing machine, the “Ruxin 1st and 2nd Generation” projects, and the “Ruyue” series washing machines, covering code testing, UI design, and machine debugging.
- Participated in system design, code implementation, debugging, technical documentation, and milestone delivery for the “Ruxin 1st and 2nd Generation” projects; the 2nd generation project received a company-level science and technology progress award, and led the upgrade of the Allwinner R311 chip display platform.
- Led the development of an emergency exit project (new series) and independently completed the overall design and development of the color display panel control firmware. Restructured and optimized the parameter storage architecture based on functional requirements, and iteratively upgraded the display logic and key-press interaction handling mechanisms. Targeted and resolved critical intermittent issues—clock divider anomalies, serial port failures, and unresponsive power-on buttons—that exhibited extremely low trigger probability, no fixed reproduction conditions, and could not be localized through simulation due to the large-scale codebase. Conducted over a thousand power-cycle tests and established program control groups for multi-dimensional comparative verification, precisely identifying the fault coupling. Verified that the main loop operated normally and that the fault was solely caused by abnormal clock configuration. By examining SCB register data and cross-referencing the chip’s official manual, uncovered a probabilistic clock-zeroing vulnerability during real-time dynamic frequency division, and determined the underlying chip-level logic requiring the main clock source to be disabled before frequency division configuration and re-enabled afterward. Completed root cause identification, solution validation, and code optimization within one month, permanently eliminating the low-level timing defect.
- Simultaneously resolved multiple operational hazards—push-rod motor actuation lag, probabilistic motor non-stop after door locking, and delayed key response upon repeated power cycling—thereby comprehensively enhancing device stability and ensuring the on-time, high-quality delivery of the urgent project.

Patent:

- **Title:** A Washing Machine and its Control Method, Device, Equipment, and Medium, **Patent Number:** CN 119287625 B, **Publication Date:** January 10, 2025
- **Title:** Method for Controlling Washing Rhythm of Laundry Equipment, Laundry Equipment, and Electronic Device, **Patent Number:** CN 117926540 B, **Publication Date:** April 26, 2024
- **Title:** A Washing Machine Control Method, a Device, and a Washing Machine, **Patent Number:** CN 112251983 B, **Publication Date:** January 22, 2021

COMPETITIONS

National University Student Data Modelling Competition

07/2017-10/2017

Member

- Analysed data for the “Photo-for-Reward” task, identified the Pearl River Delta as the operational context, and segmented geographical locations into three metropolitan areas, namely Guangzhou–Foshan, Shenzhen, and Dongguan
- Adopted the K-means clustering algorithm to analyse metropolitan areas, and divided tasks into 200 task groups to reduce average task spacing by 13 km and increase task completion rates by 11.3%
- Compared data differences between pre- and post-grouping, carried out mathematical modelling and data fitting analysis, and drafted a final design proposal

- **Achievement:** Provincial-Level Second Prize in the Shanxi Division

Shenzhen Cup Mathematical Modelling Challenge and the 4th Shanxi University Student Mathematical Modelling Competition

05/2017-07/2017

Member

- Analysed variable relationships and user evaluations about online videos and built a fitting relation about overall initial buffering delay to initial buffering peak rate and average playback rate, achieving a fitting coefficient (R^2) of 99.97%

- Listed other variables from the study, and established a functional relationship between stalling duration proportion and network-side variables, such as initial buffering peak rate, average playback-stage download rate, end-to-end round-trip time, achieving a fitting coefficient (R^2) of 96.9%
- Converted dependent variables into a five-point scoring system based on functional formulas, calculated the impact of each dependent variable on overall scores using the entropy weight method, and developed a final five-point model
- **Achievement: Provincial-Level Third Prize** in the Shanxi Division

ADDITIONAL INFORMATION

- **Languages:** Chinese; English (CET-6)
Technical Skills: C; C++; Qualification of Computer and Software Professional(Senior Systems Analyst); Microsoft Office (National Computer Rank Examination (Level 2));
- **Hobbies:** Reading; Programming; Badminton; Animation